Questions on Lecture of cell structure

Choose the correct answer:

1. **In eukaryotic cell DNA is present in:**

   a)In cytoplasm. b)**In mitochondria.** c) **nucleus** d) **None of the above**

2. **In prokaryotic cell DNA is present in:**

a) nucleoid region. b) **mitochondria.** c) **nucleus**

3. **Water constitutes ……… % of the body weight**:

   a)About 90% b)About 20% c)About 55% d) About 30%

4. Water constitutes ……… % of the body weight in men:
   a)About 90% b)About 20% c)About 60% d) About 30%

5. **Water daily intake for adults should be not less than :**

a) 10 liters b) 5 liters c)At least l.5 liters d)500 ml of water

6. **A 70 kilogram man has about 45 liters of water in his body, 30 liters of them are present in**

   a- Intracellular Fluid b- Extracellular Fluid c- Interstitial Fluid

7. **The organelle responsible for packaging in eukaryotic cells is :**
   a)Lysosome b)**Golgi apparatus**
   c)**Endoplasmic reticulum** d)**Mitochondria**

8- The organelle responsible for ATP production in eukaryotic cells is :

a)Lysosome b)**Golgi apparatus**
c)**Endoplasmic reticulum** d)**Mitochondria**

9- The organelle responsible for protein synthesis **in eukaryotic cells is :**
a)Lysosome b)**Golgi apparatus**
c)**Endoplasmic reticulum** d)**Mitochondria**
10- ………………is the main type of lipid in cell membrane.

a)phospholipids b) Glycolipids c) lipoproteins d) Glycerol

11- The hormone that maintain water at equilibrium in the body is:

a) ADH b)thyroid hormone c) insulin hormone

Compare between Prokaryotic and Eukaryotic cells

Feature	Prokaryotic Cells	Eukaryotic Cells
Genetic Material	**Circular DNA, no nucleus**	**Linear DNA, nucleus**
DNA Packaging	**No histones**	**Histone-associated chromatin**
Cell Membrane	**Phospholipids only**	**Phospholipids + sterols**
Organelles	**Absent**	**Present**
Ribosomes	**70S**	**80S**

Compare between Prokaryotic and Eukaryotic cells regarding genetic material

Prokaryotic cells	Eukaryotic cells
Circular DNA	**Linear DNA**
No histones	**DNA wrapped around histones**
Plasmid	**No Plasmid**

Match from column A to the suitable answer in column B

A	B
1- **Mitochondria** **(d)**	a- packaging
2- **Endoplasmic reticulum** **(c)**	b- digestion
3- **Golgi apparatus** **(a)**	c- protein synthesis
4- **Lysosomes** **(b)**	d- ATP synthesis

Define the following terms:

<u>Apoptosis</u>: It is a programmed cell death. It is a controlled process of cellular self-destruction

<u>Neoplasia:</u> is an abnormal accumulation of cells that occurs because of an imbalance between cellular proliferation and cellular attrition.

Complete the following space with suitable word:

1- **It is a programmed cell death. It is a controlled process of cellular self-destruction… <u>Apoptosis</u> ……….**
2- **is an abnormal accumulation of cells that occurs because of an imbalance between cellular proliferation and cellular attrition… <u>Neoplasia:</u>……….**
3- **Thin envelopes that define the volume of cells and cellular organells……<u>Membranes</u>……….**

<u>Enumerate 3 functions for the Membranes:</u>

1-Membranes constitute the boundaries of cells and intracellular organelles.

2-Membranes have proteins that mediate and regulate the transport of metabolites, macromolecules and ions.

3- Exocytosis and endocytosis

……………………………

Questions on Lecture of carbohydrate biochemistry

Choose the correct answer:

1-Which one of the following sugars is a pentose.
a-Ribulose. b- Lactose c-Mannose. d-Glucose

2-Which one of the following sugars is a pentose.

a-Ribose.b- Lactose c-Fructose. d-Glyceralhyde

3-Which one of the following sugars is found in RNA?

a-Xylose. b-Maltose c-Ribose d- Glucose

4-Which one of the following sugars is found in DNA?

a- Deoxy-ribose. b- ribose c- ribulose d-xylulose

5- Which one of the following sugars is a monosaccharide?

6-a- Sucrose. b- Glucose c- Lactose d-Maltose

7-Which one of the following sugars is a monosaccharide?

a- Sucrose. b-Fructose c- Lactose d-Maltose

8-Which one of the following sugars is a disaccharide?
a- Starch. b-Lactose c- Galactose d-Glucose

9-Which one of the following sugars is a disaccharide?
a- Glycogen. b-Sucrose c- Fructose d-Glucose

10- Which one of the following sugars is a disaccharide?

a- Starch. b-Maltose c- Ribose d-Glucose

11- Which one of the following is homopolysaccharides:

a- Heparin b-Glycogen

c- proteoglycan d-Hyaluronic acid

12- Which one ofthe followings is homopolysaccharide:

a-Keratansulphate b- Starch

c-Chondoitinsulphate d-Hyaluronic acid

13- Alpha 1-4 and alpha 1-6 glycosidic bonds are present in one of the following polysaccharides:

a- Amylose b- Glycogen c- Inulin d-Cellulose

14- Alpha 1-4 and alpha 1-6 glycosidic bonds are present in one of the following polysaccharides:

a- Amylose b- Starch

c- Hyaluronic acid d-Cellulose

15- Which of the following is a micronutrient?

a.Carbohydrates b. Fat c. Protein d.Vitamins and minerals

16- Dietary carbohydrates yield.................. energy/gram

a.4Kcal b.7Kcal c.9Kcal d.12Kcal

17- Dietary lipids yield .... energy/gram

a.4Kcal b.7Kcal c.9Kcal d.12Kcal

18- Dietary proteins yield .... energy/gram

a.4Kcal b.7Kcal c.9Kcal d.12Kcal

19- The non reducing disaccharides is:

a)Sucrose b) lactose c)fructose d) Glucose

20- The most preferable source of energy for brain is:

a) Glucose b) lipids c) fructose d) proteins

21- The only source of energy for RBCs is:

a) Glucose b) lipids c) fructose d) protiens

22- Which of the following has anticoagulant properties:

a) Heparin b) Hyaluronic acid c) Chondrotion sulfates

23- Carbohydrates that are formed of (3-10) sugar unites linked by glycosidic linkage are called:

a- Monosaccharides b- Oligosaccharides

c- Disaccharides d- Polysaccharides

24- Which one of the following is considered ketose?

a- Glucose b- Glyceraldhyde

c- Fructose d- Erythrose

25- monosaccharides linked to gether by……….to form disaccharides and poly saccharides

a) Ester bond b) glycosidic bond c) peptide bond

Complete the following space with suitable word:

1-Compounds that are needed in amounts of a gram or more per day Macro nutrients… For example, carbohydrate, protein, fat.

2- Compounds that are needed in amounts of a milligram or less per day Micro nutrients… For example, vitamins, minerals,

3- Two examples for hexoses include …glucose, fructose………

4- Examples for pentoses include,,,,ribose, deoxyribose

……………………………………………………..

Enumerate functions for the following:

A) Function of carbohydrates:

1- Glucose is the most important dietary carbohydrate and the major

metabolic fuel to the body.

2-<u>Glycogen</u> is a storage form of carbohydrates in liver and muscles

3-<u>Ribose</u> and <u>deoxyribose</u> are important for synthesis of nucleic acids (RNA&DNA)

4-<u>Galactose</u> is important for synthesis of lactose of milk.

5-<u>Fructose</u> is the fuel sugar for sperms

B) Function of pentoses (ribose and deoxyribose**)**

1- pentoses enters in the structure of DNA (deoxyribose) and RNA (ribose)

2- pentoses enters in the structure of important free nucleotides such as ATP, GTP and coenzymes NAD , NADP and FAD, FMN.

C) Function of Glycoproteins:

- **Structural function e.g. cell walls, collagen, elastin, fibrin and bone matrix.**
- **Lubricant and protective functions e.g. mucins and mucous secretions.**
- **Immunologic function e.g. immunoglobulin.**
- **Hormonal function e.g. thyroid stimulating hormone (TSH).**

Match from column A to the suitable answer in column B

A		B
1- Glucose	(d)	a- Anticoagulant
2- Fructose	(c)	b- enter in structure of RNA
3- Heparin	(a)	c- Important energy source for sperms
4- Ribose	(b)	d- Important energy source for brain
5- Lactose	(f)	e- a storage form of carbohydrates in liver and muscles
6- Gycogen	(e)	f- milk sugar

…………………………………………………………

Questions on Lecture of lipid biochemistry

<u>Choose the correct answer:</u>

1- Essential Fatty acid that should be in human diet include one of the following:

a) Palmietic acid b) stearic acid c) oleic acid d) linoleic acid

2- Essential Fatty acid that should be in human diet include one of the following:

a) Palmietic acid b) stearic acid c) oleic acid d) linolenic acid

3- Essential Fatty acid that should be in human diet include one of the following:

a) Palmietic acid b) stearic acid c) oleic acid d) arachidonic acid

4- A non essential Fatty acid include which of the following:

a) linoleic acid b) linolenic acid c) palmitic acid d) arachidonic acid

5-A non essential Fatty acid include which of the following:

a) linoleic acid b) linolenic acid c) stearic acid d) arachidonic acid

6- Which is a major constituent of lung surfactant

a) lecithin b) cholesterol c) glycerol d) Phosphaditic acid

7- Which is a major constituent of myelin sheath

a) lecithin b) cholesterol c) cephalin d) Phosphaditic acid

8- The precursor compound of phospholipids is

a) lecithin b) cholesterol c) glycerol d) Phosphaditic acid

9- The precursor compound of prostaglandins is

a) lecithin b) cholesterol c) glycerol d) arachidonic acid

10- An unsaturated fatty acid with 3(three) double bonds is

a) linoleic acid b) linolenic acid c) palmitic acid d) arachidonic acid

11- An unsaturated fatty acid with 2(two) double bonds is

a) linoleic acid b) linolenic acid c) palmitic acid d) arachidonic acid

12- An unsaturated fatty acid with 4(four) double bonds is

a) linoleic acid b) linolenic acid c) palmitic acid d) arachidonic acid

13- The number of double bonds in linolenic acid:

a)one b) two c) three d) four

14- The number of double bonds in Arachidonic acid:

a)one b) two c) three d) four

15- The number of double bonds in linoleic acid:

a)one b) two c) three d) four

16- Fatty acid linked to glycerol by……..

a) Ester bond b) glycosidic bond c) peptide bond

……………………………….….

Enumerate functions for the following:

A) Functions of lipids:

(1) An efficient source of energy (1 gm of fat about 9 cal).

(2) It serves as thermal insulators in the subcutaneous tissues and around certain organs e.g kidney

(3) Lipids act as electrical insulators, allowing rapid propagation of the impulses along the myelinated nerves.

(4) Source for Fat-soluble vitamins and essential fatty acids.

(6) Cholesterol and its important derivatives as bile acids and steroid hormones.

B) Functions of Cholesterol:

1- cholesterol is a major constituent of plasma membrane.

2-synthesis of Bile acids: There are 2 types: cholic acid and chenodeoxycholic acid.

3- synthesis of Steroid hormones:

(A)Sex hormones: estrogens, androgens and progesterone.

(B) cortisone and aldosterone.

3- synthesis of Vitamin D : synthesized in the skin from cholesterol and activated in liver and kidney.

C) **Importance of essential fatty acids**:

1- Formation of phospholipids and cholesterol ester.

2- Enter in structure of cell membrane and are required for the fluidity of cell membrane.

3-They protect against atherosclerosis and coronary heart disease.

4- Arachidonic acid is a precursor of eicosanoids (e.g. prostaglandins).

Define the following:

Rancidity: is the development of bad odor and taste in the fat

Match from column A to the suitable answer in column B

A	B
1- Rancidity (d)	**a- synthesized from** cholesterol
2- Arachidonic acid (c)	**b-** An unsaturated fatty acid with 3 double bonds
3- Steroid hormones (a)	**c-** a precursor of prostaglandins
4- linolenic acid **(b)**	**d-** development of bad odor and taste in the fat
5- Stearic acid (f)	**e- a storage form of carbohydrates in liver and muscles**

……………………………………………………………………

Questions on Lecture of protein biochemistry

Choose the correct answer:

1- The metabolizable forms of amino acids are….

a- L-aminoacids

c-R-aminoacids

b- D-aminoacids

d- None of these

2- Protein denaturation leads to all of the following effects EXCEPT:

a- Loss of catalytic activity of enzymes

b- Loss of biological functions of hormones

c- Decreased solubility

d- **Decreased digestibility**

3- All the following proteins are of high biological value EXCEPT

a- Albumin

b-Histones

c- Glutelines

d-Globulins

4- All the followings are globular proteins EXCEPT

a- Albumin

b-Globulin

c- Insulin

d-Collagen

5- During denaturation of proteins all the bonds are destroyed except:

a) The hydrogen bonds
b) The ionic bonds
c) The peptide bonds

6- High biological value proteins are:

a) Albumin and globulin

b) Protamines and glutelins

c) Globulins and histones

a) Aspartate

b) Valine

c) Phenylalanine

7- Arginine is an example of

a) **Acidic amino acids**
b) **Basic amino acids**
c) **Neutral amino acids**

8- The only amino acid that is purely ketogenic is:

a) Lysine

b) Leucine

c) Isoleucine

9- Essentialaminoacidsincludeallofthefollowingeffects<u>EXCEPT</u>

a) Aspartate
b) Tryptophan
c) Valine
d) Phenylalanine

10- **Scleroproteins include**:

a) collagen, elastin
b) collagen, albumin
c) albumin, globulins

11- **Acidic amino acids are:**

a- mono-amino,monocarboxylic

b- mono-amino,dicarboxylic

c- di-amino,monocarboxylic

Define :-

1- **<u>proteins:</u>**

2- **<u>Denaturation of proteins:</u>** It is the destruction of the secondary, tertiary and quaternary structures of a protein molecule.

3- **<u>High biological value proteins:</u>** Contain all essential amino acids, and easily digested. E.g. albumin, globulins and glutelines.

4- **Essential amino acids:** Not synthesized in the body and must be supplied in the diet.

5- **Fibrousproteins:** have an axial ratio greater than 10 . The polypeptide chains coiled in spiral or helix and cross linked by hydrogen bonds like keratin, collagen and fibrin

Enumerate:-

2- Functions of proteins:

- **Nutritional:** each 1 gram protein gives about 4 Kcal.
- **Catalytic role:** All enzymes are proteins in nature.
- **Hormonal role:** Most of hormones and all cellular receptors are protein in nature.
- **Immunological role:** The antibodies (immunoglobulins) that play an important role in the body's defensive mechanisms are proteins in nature.
- **Structural role (e.g. collagen)**
- **Plasma proteins** are responsible for most effective **osmotic pressure** of the blood

3- Functions of albumin and globulins

- **Both function as protein carrier for transport of different elements, vitamins & hormones.**
- **Albumin keeps blood osmotic pressure.**
- **Antibodies are γ-globulins.**

4- Effects of denaturation on proteins.

- **Loss of catalytic activity of enzymes**
- **Loss of biological functions of hormones 3-Decreased solubility**
- **Increased viscosity**

- **Increased digestibility e.g. cooking meat & eggs by boiling**

5- Applications of Denaturationon proteins.

- Denatured proteins, e.g., cooked meat are easily digested.
- Avoidance of denaturation is important for biological samples used for determination of enzymatic, hormonal or protein contents.
- Detection of albumin in urine by heat coagulation test is based on denaturation by heat.

6- Five of essential amino acids.

Isoleucine – Lysine- Tryptophan – Methionine - Leucine

7- Two smi- essential amino acids.

Histidine - **Arginine**

8- Metabolic classification subtype of amino acids.

a) glucogenic amino acids
b) ketogenic amino acids
c) mixed amino acids

9- Five types of neutral amino acids.

1 **Aliphatic amino acids:** e.g., Glycine, Alanine.

2 **Hydroxy amino acids:** e.g., serine, threonine, tyrosine.

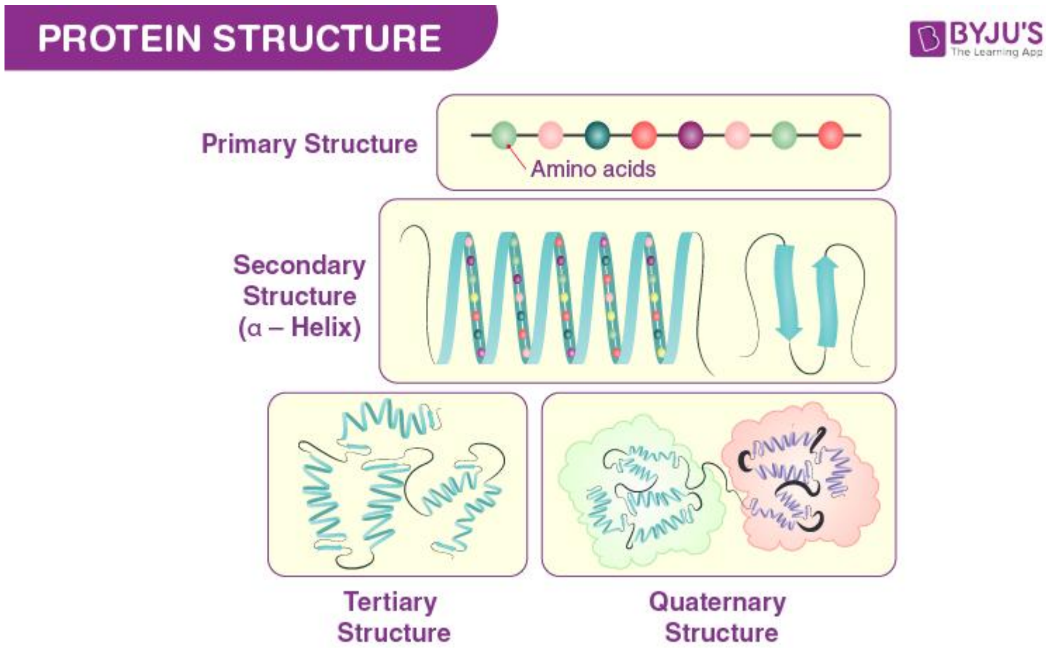
3 **Aromatic amino acids:** e.g., tyrosine, phenylalanine.

4 **Sulfur-containing amino acids:** e.g., Cysteine, homocysteine, cystine

5 **Heterocyclic amino acids: e.g.,** Tryptophan, histidine

10- Four orders of structural organization of proteins.

a. primary structure
b. secondary structure
c. tertiary structure
d- quaternary structure



Compare between Marasmus and kwashiorkor diseases**:**

	Marasmus	Kwashiorkor
edema	Absent	present
Enlarged fatty liver	Absent	present
Changes in plasma proteins	Absent	present
Skin lesions	Absent	present